

Time intelligence functions

INTERMEDIATE DAX IN POWER BI



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Time intelligence functions

- Manipulate and compare data using time periods



- Compare current period with previous period
- Estimate monthly/quarterly/yearly goals

- Many time intelligence functions exist

Time intelligence functions returning a date

- `NEXTDAY(<dates>)`
 - *Returns the next day*

dates	NEXTDAY
2009-07-07	2009-07-08
2009-07-08	2009-07-09
2009-07-09	2009-07-10

Time intelligence functions returning a date

- `NEXTDAY(<dates>)`

- *Returns the next day*

- `SAMEPERIODLASTYEAR(<dates>)`

- *Returns the last year*

dates	NEXTDAY	LASTYEAR
2009-07-07	2009-07-08	2008-07-07
2009-07-08	2009-07-09	2008-07-08
2009-07-09	2009-07-10	2008-07-09

- `DATESBETWEEN(<dates>, <start_date>, <end_date>)`

- *Returns dates between start and end date*

Time intelligence functions returning a date

- `NEXTDAY(<dates>)`
 - *Returns the next day*
- `SAMEPERIODLASTYEAR(<dates>)`
 - *Returns the last year*
- `DATESBETWEEN(<dates>, <start_date>, <end_date>)`
 - *Returns dates between start and end date*

dates	NEXTDAY	LASTYEAR
2009-07-07	2009-07-08	2008-07-07
2009-07-08	2009-07-09	2008-07-08
2009-07-09	2009-07-10	2008-07-09

dates	DATESBETWEEN
2009-07-07	
2009-07-08	2009-07-08
2009-07-09	2009-07-09
2009-07-10	

Time intelligence functions returning a date

```
Mid Season Sales =  
CALCULATE(  
    SUM(Fact_Table[Sales]),  
    DATESBETWEEN(Dim_Date[Date Key],  
        DATE(2014, 10, 04),  
        DATE(2014, 10, 26)  
    )  
)
```

Time intelligence functions returning a date

```
TOTALYTD(<expression>, <dates> [,<filter>])  
TOTALQTD(<expression>, <dates> [,<filter>])  
TOTALMTD(<expression>, <dates> [,<filter>])
```

Returns the year, quarter, or month to date value of the expression.

```
Sum_YTD =  
TOTALYTD(  
    SUM(Fact_Table[Value]),  
    Dim_Date[Date Key]  
)
```


Time intelligence functions returning a date

```
TOTALYTD(<expression>, <dates> [,<filter>])  
TOTALQTD(<expression>, <dates> [,<filter>])  
TOTALMTD(<expression>, <dates> [,<filter>])
```

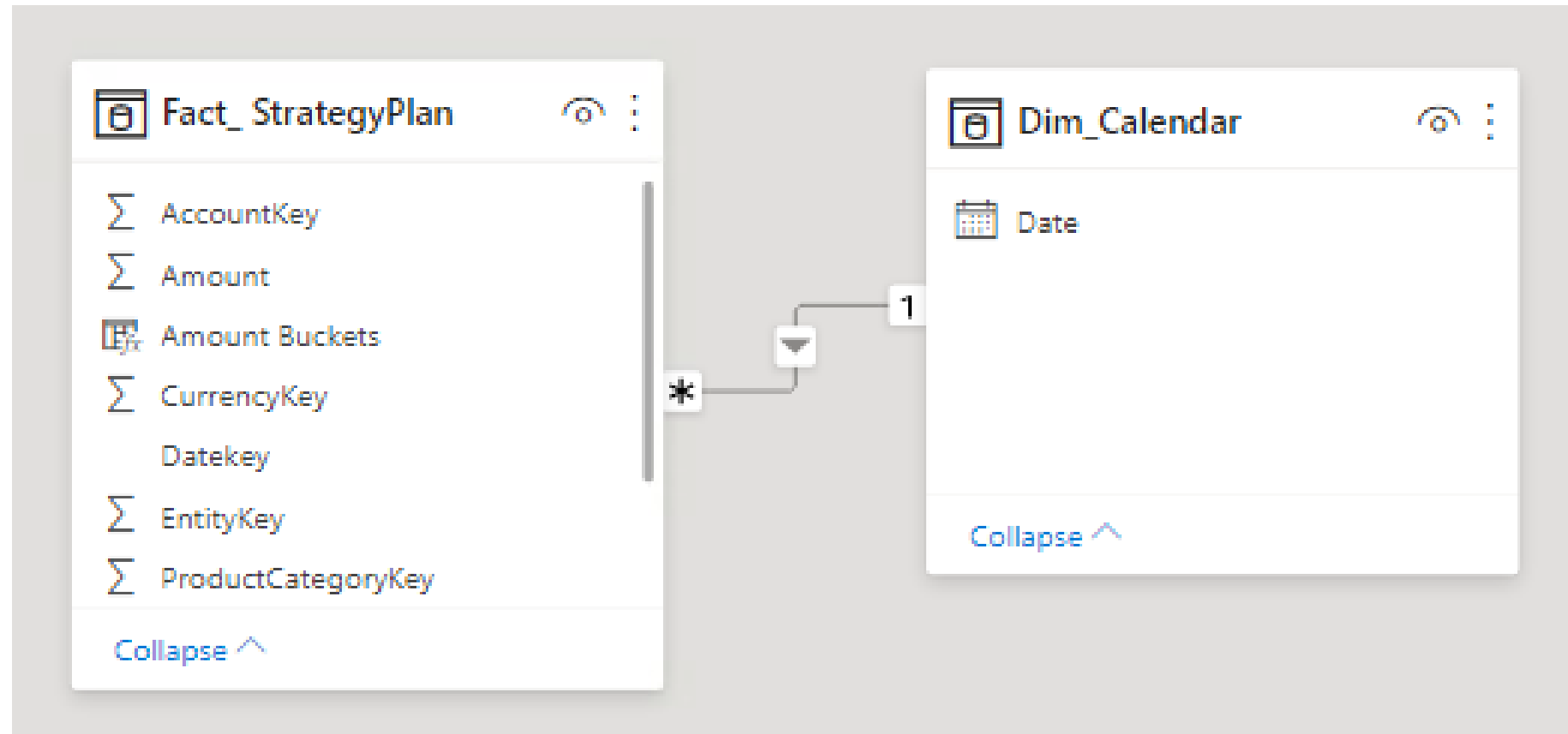
Returns the year, quarter, or month to date value of the expression.

```
Sum_YTD =  
TOTALYTD(  
    SUM(Fact_Table[Orders]),  
    Dim_Date[Date Key]  
)
```

Year	Month	Value	Sum_YTD
2021	Jan	6,532	6,532
2021	Feb	4,263	10,795
2021	Mar	1,256	12,051
Total		12,051	12,051

Best practices for time intelligence functions

- Use a separate date dimension table



A date column in the fact table could contain missing dates!

Let's practice!

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Time intelligence in Power BI

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Let's practice!

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Congratulations!

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What you've learned so far

Chapter 1

Logical functions:

- `IF()`
- `SWITCH()`

Chapter 3

Table manipulation functions

- `ADDCOLUMNS()`
- `SUMMARIZE()`

Chapter 2

Row-level security

- `USERPRINCIPALNAME()`

Chapter 4

Time intelligence functions

- `SAMEPERIODLASTYEAR()`
- `TOTALYTD()`

Use it or lose it

Download our cheat sheet on the course landing page!

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Glossary of DAX

functions and operators

> Math & statistical functions

- SUM(<column>)**
Adds all the numbers in a column.
- AVERAGE(<column>)**
Returns the average (arithmetic mean) of all the numbers in a column.
- SUMX(<table>, <expression>)**
Returns the sum of an expression evaluated for each row in a table.
- COUNTX(<table>, <expression>)**
Counts the number of rows from an expression that evaluates to a non-blank value.
- AVERAGEX(<table>, <expression>)**
Calculates the average (arithmetic mean) of a set of expressions evaluated over a table.
- DIVIDE(<numerator>, <denominator> [, <alternateresult>])**
Performs division and returns alternate result or BLANK() on division by 0.
- MIN(<column>)**
Returns a minimum value of a column.
- MAX(<column>)**
Returns a maximum value of a column.
- COUNTROWS(<table>)**
Counts the number of rows in a table.
- DISTINCTCOUNT(<column>)**
Counts the number of distinct values in a column.
- RANKX(<table>, <expression>[, <value>[, <order>[, <ties>]]])**
Returns the ranking of a number in a list of numbers for each row in the table argument.

> Filter functions

- FILTER(<table>, <filter>)**
Returns a table that is a subset of another table or expression.
- CALCULATE(<expression>[, <filter>] [, <filter2>] [, ...]))**
Evaluates an expression in a filter context.
- HASONEVALUE(<columnName>)**
Returns TRUE when the context for columnName has been filtered down to one distinct value only. Otherwise it is FALSE.
- ALL(<table> | <column>[, <column>[, <column>[,...]]])**
Returns all the rows in a table, or all the values in a column, ignoring any filters that might have been applied.

> Logical functions

- IF(<logical_test>, <value_if_true>[, <value_if_false>])**
Checks a condition, and returns a certain value depending on whether it is true or false.
- AND(<logical 1>, <logical 2>)**
Checks whether both arguments are TRUE, and returns TRUE if both arguments are TRUE. Otherwise, it returns FALSE.
- OR(<logical 1>, <logical 2>)**
Checks whether one of the arguments is TRUE to return TRUE. The function returns FALSE if both arguments are FALSE.
- NOT(<logical>)**
Changes TRUE to FALSE and vice versa.
- SWITCH(<expression>, <value>, <result>[, <value>, <result>]...[, <else>])**
Evaluates an expression against a list of values and returns one of multiple possible result

> Date & time functions

- CALENDAR(<start_date>, <end_date>)**
Returns a table with a single column named "Date" that contains a contiguous set of dates.

> Time intelligence functions

- TOTALYTD(<expression>, <dates>[, <filter>][, <year_end_date>])**
Evaluates the year-to-date value of the expression in the current context.
- SAMEPERIODLASTYEAR(<dates>)**
Returns a table that contains a column of dates shifted one year back in time.

> Relationship functions

- CROSSFILTER()**
Specifies the cross-filtering direction to be used in a calculation.
- RELATED()**
Returns a related value from another table.

> Table manipulation functions

- SUMMARIZE(<table>, <groupBy_columnName>[, <groupBy_columnName>]...[, <name>, <expression>]...)**
Returns a summary table for the requested totals over a set of groups.
- DISTINCT(<table>)**
Returns a table by removing duplicate rows from another table or expression.
- ADDCOLUMNS(<table>, <name>, <expression>[, <name>, <expression>]...)**
Adds calculated columns to the given table or table expression.
- SELECTCOLUMNS(<table>, <name>, <expression>[, <name>, <expression>]...)**
Selects calculated columns from the given table or table expression.

> Text functions

- SUBSTITUTE(<text>, <old_text>, <new_text>, <instance_num>)**
Replaces existing text with new text in a string.

> Information functions

- USERPRINCIPALNAME()**
Returns the user principal name or email address. This function has no arguments.

> DAX statements

- VAR(<name> = <expression>)**
Stores the result of an expression as a named variable. To return the variable, use RETURN after the variable is defined.

> Other functions

- BLANK()**
Returns a blank.

> DAX Operators

Comparison operators	Meaning
=	Equal to
= =	Strict equal to
>	Greater than
<	Smaller than
> =	Greater than or equal to
= <	Smaller than or equal to
< >	Not equal to

Text operator	Meaning	Example
&	Concatenates text values	Concatenates text values [City]&" [State]"

Logical operator	Meaning	Example
&&	AND condition	([City] = "Brg") && ([Return] = "Yes")
	OR condition	([City] = "Brg") ([Return] = "Yes")
IN { }	OR condition for each row	Product[Color] IN {"Red", "Blue", "Gold"}

Can't find the function you're looking for?
Take a look at the Microsoft [documentation](#)

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See you again soon!

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